Android Assessment (One Day Assignment)

O3 Interfaces

In this assignment, you are expected to develop a generic function for Android/iOS mobile phones to calculate the cost of electricity consumption based on the electric meter readings. In typical cases, electricity provider companies devise slabs and rates per slab such as follows:

Consumption	Rate
1-100 units	@ $5 / unit
101 – 500 units	@ $8 / unit
> 500 units	@ $10 / unit

The slab table above is just given as an illustration. In your work, we expect you to implement a configurable slab table where we can define any number of slabs with different consumption units and rates.

The cost of consumption for a household is calculated by finding the difference between current electric meter reading and previous electric meter reading for this household and finally applying the rates given on the slab table. Here are some examples to show how it is done:

1- Assume that a particular household **(HA123)** electric meter reading is 250 units, and this is the first reading of this household, which means there are no previous readings, then the cost is calculated according to the slab configuration as follows:

- 100 units x 5 = $500 (First 100 units will cost $5/unit)
- 150 units x 8 =$1,200 (Remaining 150 units fall into the 2(nd) row in the slab table)

Total Bill Amount = $500 + $1,200 = $1,700

2- We have a second household **(HA456)** and assume the current electric meter reading for this household is 510 units and this household has no previous readings, then the cost is calculated as:

- 100 units x 5 = $500 (1(st) row of slab table)

- 400 units x 8 = $3,200 (2(nd) row of the slab table)
- 10 units x 10 = $100 (3(rd) row of the slab table)

Total Bill Amount = $3,800

3- Later we got a new reading for the household (**HA456**). The reading shows 630 units. We should now find the difference between the previous reading and current reading and find the actual consumption:

- Difference between readings is 630 (current reading) - 510 (previous reading) = 120
- The actual consumption is 120 units, so we calculate the cost as follows:

o 100 units x 5 = $500 (1(st) row in slab table)

o 20 units x 8 = $160 (2(nd) row in slab table)

Total Bill Amount = $660

We are expecting you to develop an application which calculates electricity consumption cost for different households. The requirements are as follows:

1) Develop a screen for contents:

a. Service Number of the customer (Text box, mandatory to be 10 digits alpha numeric).

i. Screen should validate the customer service number and show appropriate validation message for incorrect values.

b. Current meter reading (Text box only positive numeric values)

i. Screen should validate the meter reading and show appropriate validation message for incorrect values.

c. Submit button

2) When clicked on “Submit” button, the application should do the following:

a. If there are previous historical readings for the customer

i. Show the last 3 historical readings and calculated costs in a table.

ii. Calculate the difference between the current reading which was entered in the text box and previous reading from history and use the slab table configuration to calculate the cost of consumption

iii. Show the consumption cost

b. If there are no previous readings for the customer or if the customer is a new customer:

i. Calculate the cost of consumption by using the reading value which was entered in the text box and using the slab table configuration

ii. Show the consumption cost

3) In the same screen after showing the consumption cost, there should be a "Save" button and when clicked on that button, the reading and the calculated cost should be stored in the historical readings for that household and the screen should return to the entry screen in the first step.

4) A new electric meter reading must always be higher (or equal) than the previous meter reading for the same customer, so new readings always increase and must not decrease. Your application should show appropriate warning messages for the following cases:

a. When a reading is less than a previous reading

b. When a reading is a negative value

5) Slab table should be configurable. Before testing your application, we are going to first configure this slab table and then test your application accordingly. You do not need to develop a slab configuration screen, but you need to define how we can configure the slabs.

How will you be evaluated?

Completeness of the solution according to the requirements given above

1) Efficiency of the algorithm to calculate cost

2) Professional look and feel of the user interface

3) Code Quality (Exception handling, validations, logging etc.)

4) Delivery of Complete Code & Application Screenshots